

SIMATS ENGINEERING

ASSESSMENT TOOL – 2

COURSE NAME: Deep Learning

COURSE CODE: MLA0406

COURSE OUTCOME COVERED (CO5)

CO5: Compare and assess emerging technologies including AI/ML and Blockchain with respect to their applicability, societal impact, ethical considerations, and sustainability in the context of modern software system design and development. (BL5)

ANNEXURE A

ERROR ANALYSIS TASK

Code of Conduct

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I certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Paleru Dharani Govardhan*

Confusion Matrix

Actual \ Predicted	Cat	Dog	Bird	Rabbit
Cat	42	5	2	1
Dog	6	38	3	3
Bird	1	4	43	2
Rabbit	2	3	5	40

Total Test Images = 200

1. Calculate the Overall Accuracy of the Model

Formula

$$[=]$$

Correct predictions are the diagonal values of the confusion matrix.

$$\text{Correct Predictions} = 42 + 38 + 43 + 40 = \mathbf{163}$$

$$\text{Total Predictions} = \mathbf{200}$$

$$[=]$$

$$[= 81.5\%$$

Answer

$$\text{Overall Accuracy} = \mathbf{81.5\%}$$

2. Identify Which Class Has the Highest Number of Misclassifications

Each class contains 50 images.

Cat

$$\text{Misclassified} = 50 - 42 = \mathbf{8}$$

Dog

$$\text{Misclassified} = 50 - 38 = \mathbf{12}$$

Bird

$$\text{Misclassified} = 50 - 43 = \mathbf{7}$$

Rabbit

$$\text{Misclassified} = 50 - 40 = \mathbf{10}$$

Class	Misclassified Images
Cat	8
Dog	12
Bird	7
Rabbit	10

Answer

The **Dog** class has the highest number of misclassifications with **12 images**.

3. List the Two Most Common Classification Errors

From the confusion matrix:

- Cat $\rightarrow$ Dog = 5
- Dog $\rightarrow$ Cat = 6
- Bird $\rightarrow$ Dog = 4
- Rabbit $\rightarrow$ Bird = 5

Answer

The two most common classification errors are:

1. **Dog classified as Cat (6 images).**
2. **Cat classified as Dog (5 images).**

Note: Rabbit classified as Bird (5 images) is also a frequent error.

4. Suggest Three Possible Reasons Why These Errors Occurred

Reason 1: Similar Visual Features

Cats and dogs have similar shapes, fur textures, and facial features, making them difficult for the CNN to distinguish.

Reason 2: Limited or Imbalanced Training Data

If certain classes have fewer or less diverse training images, the model may not learn enough distinguishing features.

Reason 3: Poor Image Quality

Low resolution, poor lighting, background clutter, different camera angles, or partially visible animals can reduce prediction accuracy.

5. Recommend Four Techniques to Improve Model Performance

1. Data Augmentation

Increase dataset diversity using image transformations such as rotation, flipping, zooming, cropping, and brightness adjustments.

2. Collect More Training Data

Add more labeled images for all classes, especially those with higher misclassification rates.

3. Use Transfer Learning

Train the model using pre-trained CNN architectures such as:

- ResNet50
- EfficientNet
- VGG16
- MobileNet

These models provide better feature extraction and often improve accuracy.

4. Hyperparameter Tuning

Optimize important parameters such as:

- Learning Rate
- Batch Size
- Number of Epochs
- Optimizer (Adam or SGD)

to improve training performance.

6. Explain Whether Collecting More Training Data or Increasing the Number of CNN Layers Would Be More Beneficial

Answer

Collecting **more training data** would be more beneficial than increasing the number of CNN layers.

Justification

- The confusion matrix indicates that the model struggles mainly with distinguishing visually similar classes rather than lacking model complexity.
- More diverse training images help the model learn robust and generalized features.
- Increasing the number of layers without sufficient data can lead to overfitting.
- A larger and more balanced dataset improves the model's ability to classify unseen images correctly.

Conclusion

Collecting more high-quality training data is the better approach because it improves generalization and reduces classification errors.

7. If the Model Achieves 99% Training Accuracy but Only 82% Testing Accuracy, What Problem Does It Suffer From?

Answer

The model suffers from **Overfitting**.

Explanation

Overfitting occurs when the CNN learns the training data too well, including noise and unnecessary details, but fails to generalize to new unseen data.

Characteristics include:

- Very high training accuracy.
- Significantly lower testing accuracy.
- Poor performance on new images.

Methods to Reduce Overfitting

1. Apply data augmentation.
 2. Add dropout layers.
 3. Use L2 regularization.
 4. Apply early stopping.
 5. Collect more training data.
 6. Reduce model complexity if necessary.
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Conclusion

The CNN model achieved an **overall accuracy of 81.5%**, showing reasonably good performance. The **Dog** class had the highest number of misclassifications, while the most common mistakes occurred between **Dogs and Cats**, mainly due to their similar visual characteristics. The model's performance can be improved by collecting more training data, applying data augmentation, using transfer learning, and tuning hyperparameters. Since the model achieves **99% training accuracy but only 82% testing accuracy**, it clearly suffers from **overfitting**, which can be reduced using dropout, regularization, early stopping, and a larger, more diverse dataset.